Key features of quadratic functions

1

a 75 **b** 75

2

a 4 **b** 273 **c** 420 **d** 0

3

a Yes **b** No **c** Yes **d** No

e Yes **f** No **g** No **h** No

i No

4

a No **b** Yes **c** No **d** Yes

e No **f** No **g** Yes **h** Yes

i Yes **j** No **k** Yes **l** No

m No **n** No **o** Yes

5

a No **b** Yes **c** No **d** No

6

a Quadratic **b** Linear **c** Quadratic **d** Neither

e Neither

7

a

i $(4,0),(-2,0)$ **ii** $4,-2$

b

i $(2,0),(-4,0)$ **ii** $2,-4$

c

i $(3,0)$ **ii** 3

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a

i $-6, 2$ **ii** -12

iii Minimum value, -16

b

i $0, 4$ **ii** 0

iii Maximum value, 4

c

i 3 **ii** 9

iii Minimum value, 0

9

a -5 **b** -25 **c** 0 **d** $x < -5$

10

a $(-4, 2)$ **b** Minimum **c** 2

11

a Minimum point

b

i No **ii** Yes **iii** No **iv** Yes

12

a Maximum point.

b

i Yes **ii** No **iii** No **iv** Yes

13 $x < 2$

14

a

b $2, -2$

x	-3	-2	-1	0	1	2	3
y	5	0	-3	-4	-3	0	5

c $(0, -4)$ **d** Minimum

e Positive

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a

i $(3, 2)$ ii Maximum iii 2

b

i $(4, 5)$ ii Minimum iii 5

c

i $(-4, -3)$ ii Maximum iii -3

16

a Quadratic

b 9

c The height from which the soccer ball is kicked

d 3

e The time at which the soccer ball hits the ground

17

a

i Minimum, -4 ii Domain: all real x, Range: $y \geq -4$

b

i Maximum, 16 ii Domain: all real x Range: $y \leq 16$

c

i Minimum, 0 ii Domain: $x > 7$ Range: $y \geq 0$

d

i Minimum, 0 ii Domain: $x < -4$ Range: $y \geq 0$

18

a

i Minimum, -1 ii $y \geq -1$

b

i Maximum, -2 ii $y \leq -2$

c

i Minimum, -3 ii $y \geq -3$

d

i Maximum, -3 ii $y \leq -3$

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